Score	IB Grade

Name: ..... **Quiz 2** **Mme Doux** *12th IB*

EXERCISE 1 **[12 marks]**

Fifteen students were weighed and their pulse rates were measured:

Weight (x kg)	46	37	32	57	47	64	42	30	52	56	65	43	36	28	40
Pulse rate (y beats per min)	65	59	54	74	69	87	61	59	70	69	75	60	56	53	58

(a) Draw a scatter diagram for the data. **[2]**



(b) Calculate r. **[1]**

(c) Describe the relationship between *weight* and *pulse rate*. **[2]**

(d) Calculate the mean point $(\overline{x}, \overline{y})$. **[2]**

(e) Locate the mean point on the scatter diagram, then use it in drawing a line of best fit. **[2]**

(f) Estimate the pulse rate of a 50 kg student. Comment on the reliability of your estimate. **[3]**

EXERCISE 2 **[14 marks]**

The prices of petrol and the number of customers per hour for sixteen petrol stations are:

Petrol price (x cents per litre)	105.9	106.9	109.9	104.5	104.9	111.9	110.5	112.9
Number of customers (y)	45	42	25	48	43	15	19	10

Petrol price (x cents per litre)	107.5	108.0	104.9	102.9	110.9	106.9	105.5	109.5
Number of customers (y)	30	23	42	50	12	24	32	17

(a) Calculate Pearson's product-moment correlation coefficient for the data. **[1]**

(b) Describe the relationship between the *petrol price* and the *number of customers*. **[2]**

(c) Use technology to find the equation of the regression line. **[2]**

(d) State and interpret the gradient of the regression line. **[3]**

(e) Estimate the number of customers per hour for a station which sells petrol at 115.9 per litre. **[2]**

(f) Estimate the petrol price at a petrol station which has 40 customers per hour. **[2]**

(g) Comment on the reliability of your estimates in **(e)** and **(f)**. **[2]**

EXERCISE 3 **[14 marks]**

Tomatoes are sprayed with a pesticide-fertiliser mix. The table below shows the *yield of tomatoes* per bush for various *spray concentrations*.

Spray concentration (x mL per L)	3	5	6	8	9	11	15
Yield of tomatoes per bush (y)	67	90	103	120	124	150	82

(a) Determine the value of r and interpret your answer. **[2]**

(b) Is there an outlier present that is affecting the correlation? **[1]**

(c) The outlier was found to be a recording error. Remove the outlier from the data set, and recalculate r. Is it reasonable to now fit a linear model? **[3]**

(d) Determine the equation of the regression line of y against x. **[2]**

(e) Use your line to estimate:

(i) the yield if the spray concentration is 7 mL per L **[2]**

(ii) the spray concentration if the yield is 200 tomatoes per bush. **[2]**

(f) Comment on the reliability of your estimates in **(e)**. **[2]**

END !